

Derivative

$$f(x) = (3x^2 - \cos(x))^5$$

$$\Rightarrow f'(x) = 5(3x^2 - \cos(x))^4 \cdot (6x + \sin(x))$$

$$h(x) = \frac{\sin(3x)}{x^2} \quad \begin{array}{l} u = \sin(3x) \quad , \quad u' = 3\cos(3x) \\ v = x^2 \quad , \quad v' = 2x \end{array}$$

$$h'(x) = \frac{3\cos(3x)(x^2) - 2x(\sin(3x))}{(x^2)^2}$$

Simplify:

$$\text{Let } N = 3\cos(3x)(x^2) - 2x(\sin(3x))$$

$$N = ~~3x^2 \cos(3x) - 2x \sin(3x)~~ 3x^2 \cdot \cos(3x) - 2x \cdot \sin(3x)$$

$$h'(x) = \frac{3x^2 \cdot \cos(3x) - 2x \cdot \sin(3x)}{x^4}$$



$$g(x) = \sqrt{\frac{x}{\cos(x)}}$$

$$\text{Let } w = \frac{x}{\cos(x)}$$

$$u = x, \quad u' = 1$$

$$v = \cos(x), \quad v' = -\sin(x)$$

$$w' = \frac{\cos(x) + x \sin(x)}{(\cos(x))^2}$$

$$g'(x) = \frac{w'}{2\sqrt{w}}$$

$$g'(x) = \frac{\frac{\cos(x) + x \sin(x)}{(\cos(x))^2}}{2\sqrt{\frac{x}{\cos(x)}}}$$

Simplify:

~~$$g'(x) = \frac{\frac{x \sin(x)}{\cos(x)}}{2\sqrt{\frac{x}{\cos(x)}}}$$~~

$$g'(x) = \frac{x \sin(x)}{\cos^2(x) \cdot 2\sqrt{\frac{x}{\cos(x)}}}$$

